

CARBOHYDRATE PRACTICE QUESTION

1. What is the general empiric formula for many simpler carbohydrates?

- A. $(C_2HO)_n$
- B. $(CH_2O)_n$
- C. $(CHO_2)_n$
- D. $(C_2H_2O)_n$

Answer: B. $(CH_2O)_n$

Explanation: Many simple carbohydrates (monosaccharides) have the empirical formula $(CH_2O)_n$, where n is typically 3, 5, or 6 (e.g., glucose is $C_6H_{12}O_6$). This reflects a hydrated carbon structure.

2. Which term describes carbohydrates with a keto group as their most oxidized functional group?

- A. Aldoses
- B. Ketoses
- C. Polyols
- D. Uronic acids

Answer: B. Ketoses

Explanation: Ketoses are carbohydrates in which the most oxidized functional group is a ketone ($C=O$) group, typically at carbon 2 (e.g., fructose).

3. A monosaccharide containing six carbons and an aldehyde group is classified as a:

- A. Ketohexose
- B. Aldopentose
- C. Aldohexose
- D. Ketopentose

Answer: C. Aldohexose

Explanation: "Aldo-" indicates an aldehyde group, and "-hexose" indicates six carbons. Thus, an aldohexose (e.g., glucose) fits this description.

4. Sugars that have the same chemical formula but differ in structure are called:

- A. Isomers
- B. Isotopes
- C. Isozymes
- D. Polymers

Answer: A. Isomers

Explanation: Isomers are compounds with the same molecular formula but different structural arrangements. Sugars like glucose and fructose are isomers ($C_6H_{12}O_6$).

5. Which sugar is the C-4 epimer of glucose?

- A. Mannose
- B. Fructose
- C. Galactose
- D. Ribose

Answer: C. Galactose

Explanation: An epimer differs at only one chiral carbon. Galactose is the C-4 epimer of glucose (they differ in the –OH orientation at carbon 4).

6. Which sugar is the C-2 epimer of glucose?

- A. Galactose
- B. Mannose
- C. Xylulose
- D. Sorbitol

Answer: B. Mannose

Explanation: Mannose differs from glucose only at carbon 2, making it the C-2 epimer of glucose.

7. Pairs of sugars that are non-superimposable mirror images are called:

- A. Anomers
- B. Epimers
- C. Enantiomers
- D. Diastereomers

Answer: C. Enantiomers

Explanation: Enantiomers are stereoisomers that are non-superimposable mirror images of each other (e.g., D- and L-glyceraldehyde).

8. In biological systems, the vast majority of sugars exist in which isomeric form?

- A. L-form
- B. D-form
- C. Trans-form
- D. Meso-form

Answer: B. D-form

Explanation: Naturally occurring sugars in living organisms are predominantly in the D-configuration (e.g., D-glucose, D-fructose).

9. Cyclization of an aldose creates a new asymmetric center at which carbon?

- A. Carbon 1
- B. Carbon 2
- C. Carbon 5
- D. Carbon 6

Answer: A. Carbon 1

Explanation: The aldehyde carbon (carbon 1) becomes chiral upon cyclization, forming a new asymmetric center (the anomeric carbon).

10. The new asymmetric center created by cyclization is designated as the:

- A. Epimeric carbon
- B. Enantiomeric carbon
- C. Anomeric carbon
- D. Alpha carbon

Answer: C. Anomeric carbon

Explanation: The newly formed chiral center at carbon 1 (or carbon 2 for ketoses) is called the anomeric carbon.

11. A six-membered sugar ring is specifically referred to as a:

- A. Furanose
- B. Pyranose
- C. Pentose
- D. Hexose

Answer: B. Pyranose

Explanation: A six-membered ring in sugars is called a pyranose (similar to the ring structure of pyran). Glucose forms a pyranose ring.

12. Which configuration is assigned when the hydroxyl group on the anomeric carbon is "down" in a Haworth projection of glucose?

- A. Alpha (α)
- B. Beta (β)
- C. Delta (δ)
- D. Gamma (γ)

Answer: A. Alpha (α)

Explanation: In the Haworth projection for D-glucose, the α -anomer has the $-OH$ on the anomeric carbon pointing down (opposite to the CH_2OH group).

13. The spontaneous interconversion between alpha and beta anomeric forms is known as:

- A. Isomerization
- B. Epimerization
- C. Mutarotation
- D. Condensation

Answer: C. Mutarotation

Explanation: Mutarotation is the spontaneous change in optical rotation as α and β anomers interconvert via the open-chain form in solution.

14. Which test is used to detect reducing sugars in urine via copper reduction?

- A. Glucose oxidase test
- B. Clinitest
- C. Gram stain
- D. Hydrogen breath test

Answer: B. Clinitest

Explanation: Clinitest uses copper reduction (Cu^{2+} to Cu^+) to detect reducing sugars in urine, based on Benedict's principle.

15. Which of the following is a non-reducing sugar because its anomeric carbons are linked?

- A. Glucose
- B. Lactose

- C. Sucrose
- D. Maltose

Answer: C. Sucrose

Explanation: Sucrose has its anomeric carbons (glucose C1 and fructose C2) linked together, so neither can open to reduce copper—making it non-reducing.

16. Oligosaccharides generally contain how many monosaccharide units?

- A. 2
- B. 3 to 10
- C. 10 to 50
- D. Over 100

Answer: B. 3 to 10

Explanation: Oligosaccharides are defined as polymers containing 3 to 10 monosaccharide units.

17. Polysaccharides can be hundreds of sugar units in length and are often:

- A. Only linear
- B. Only branched
- C. Either linear or branched
- D. Always cyclic

Answer: C. Either linear or branched

Explanation: Polysaccharides such as amylose (linear) and glycogen (branched) show that they can be either form.

18. The primary storage polysaccharide in animals is:

- A. Starch
- B. Cellulose
- C. Glycogen
- D. Amylopectin

Answer: C. Glycogen

Explanation: Glycogen is the main glucose storage polysaccharide in animals, stored in liver and muscle.

19. The primary storage polysaccharide in plants is:

- A. Glycogen
- B. Starch
- C. Chitin
- D. Hyaluronic acid

Answer: B. Starch

Explanation: Starch (amylose and amylopectin) is the primary energy storage polysaccharide in plants.

20. Which plant polysaccharide is composed of unbranched $\beta(1\rightarrow4)$ linkages?

- A. Amylose
- B. Amylopectin

C. Cellulose

D. Glycogen

Answer: C. Cellulose

Explanation: Cellulose is an unbranched polymer of glucose linked by $\beta(1\rightarrow4)$ glycosidic bonds, providing structural support in plant cell walls.

Part II: Glycosidic Bonds and Complex Sugars

21. What type of bond joins two monosaccharides?

A. Peptide bond

B. Phosphodiester bond

C. Glycosidic bond

D. Ester bond

Answer: C. Glycosidic bond

Explanation: A glycosidic bond is the covalent linkage formed between the anomeric carbon of one monosaccharide and another molecule (often another sugar).

22. All sugar-to-sugar glycosidic bonds are of which type?

A. N-type linkages

B. O-type linkages

C. S-type linkages

D. Hydrogen linkages

Answer: B. O-type linkages

Explanation: Sugar-to-sugar bonds are O-glycosidic linkages, as they connect through an oxygen atom (between anomeric carbon and a hydroxyl group).

23. A glycosidic bond between a sugar and an $-NH_2$ group is called a:

A. O-glycoside

B. N-glycoside

C. Amide bond

D. Glycoprotein link

Answer: B. N-glycoside

Explanation: An N-glycosidic bond connects a sugar to a nitrogen-containing group (e.g., in nucleosides or glycoproteins).

24. Which enzyme type forms glycosidic bonds using nucleotide sugars like UDP-glucose?

A. Glycosidases

B. Glycosyltransferases

C. Kinases

D. Isomerases

Answer: B. Glycosyltransferases

Explanation: Glycosyltransferases catalyze the transfer of sugar moieties from activated nucleotide sugars (like UDP-glucose) to form glycosidic bonds.

25. Lactose is composed of glucose and which other monosaccharide?

A. Fructose

- B. Galactose
- C. Mannose
- D. Ribose

Answer: B. Galactose

Explanation: Lactose (milk sugar) is a disaccharide of glucose and galactose.

26. The specific linkage in lactose is:

- A. $\alpha(1 \rightarrow 4)$
- B. $\alpha(1 \rightarrow 6)$
- C. $\beta(1 \rightarrow 4)$
- D. $\alpha(1 \rightarrow 2)$

Answer: C. $\beta(1 \rightarrow 4)$

Explanation: Lactose has a $\beta(1 \rightarrow 4)$ glycosidic bond between galactose and glucose.

27. Sucrose is a disaccharide of glucose and:

- A. Galactose
- B. Fructose
- C. Glucose
- D. Xylulose

Answer: B. Fructose

Explanation: Sucrose (table sugar) is composed of glucose and fructose linked via their anomeric carbons.

28. Maltose consists of two glucose units linked by which bond?

- A. $\alpha(1 \rightarrow 4)$
- B. $\alpha(1 \rightarrow 6)$
- C. $\beta(1 \rightarrow 4)$
- D. $\beta(1 \rightarrow 2)$

Answer: A. $\alpha(1 \rightarrow 4)$

Explanation: Maltose is a disaccharide of two glucose molecules linked by an $\alpha(1 \rightarrow 4)$ glycosidic bond.

29. Which sugar is found exclusively in milk and dairy products?

- A. Sucrose
- B. Maltose
- C. Lactose
- D. Starch

Answer: C. Lactose

Explanation: Lactose is the primary sugar in milk and dairy products and is not found naturally in other common foods.

30. High-fructose corn syrup typically contains approximately what percentage of fructose?

- A. 10%
- B. 55%

- C. 90%
- D. 100%

Answer: B. 55%

Explanation: High-fructose corn syrup (HFCS-55) commonly contains about 55% fructose and 45% glucose.

31. Amylopectin and glycogen contain which type of bond at their branch points?

- A. $\alpha(1 \rightarrow 4)$
- B. $\alpha(1 \rightarrow 6)$
- C. $\beta(1 \rightarrow 4)$
- D. $\alpha(1 \rightarrow 2)$

Answer: B. $\alpha(1 \rightarrow 6)$

Explanation: Branch points in amylopectin and glycogen are formed by $\alpha(1 \rightarrow 6)$ glycosidic bonds.

32. Proteins that have sugars attached to them are called:

- A. Glycolipids
- B. Glycoproteins
- C. Proteolipids
- D. Lipoproteins

Answer: B. Glycoproteins

Explanation: Glycoproteins are proteins with covalently attached carbohydrate chains (e.g., many cell surface receptors).

33. Sugar alcohols, formed by the reduction of an aldehyde group, are also called:

- A. Aldonic acids
- B. Polyols
- C. Ketoses
- D. Uronic acids

Answer: B. Polyols

Explanation: Sugar alcohols (e.g., sorbitol, mannitol) are also known as polyols, formed by reducing the aldehyde or ketone group.

34. Glucose can be reduced to which sugar alcohol, which is associated with diabetic complications?

- A. Galactitol
- B. Sorbitol
- C. Mannitol
- D. Xylitol

Answer: B. Sorbitol

Explanation: Glucose is reduced to sorbitol by aldose reductase; sorbitol accumulation in tissues (e.g., lens) contributes to diabetic complications.

35. The reduced form of galactose that accumulates in galactosemia is:

- A. Sorbitol

- B. Galactitol
- C. Dulcitol
- D. Lactitol

Answer: B. Galactitol

Explanation: In galactosemia, galactose is reduced to galactitol, which accumulates in tissues like the lens and causes cataracts.

36. Which complex consists of a core protein with many long GAG chains attached?

- A. Glycoprotein
- B. Proteoglycan
- C. Glycolipid
- D. Peptidoglycan

Answer: B. Proteoglycan

Explanation: Proteoglycans have a core protein with multiple glycosaminoglycan (GAG) chains attached, forming large complexes in extracellular matrix.

37. The glycosidic bond in sucrose connects which carbons?

- A. α -1 of glucose to 4 of fructose
- B. α -1 of glucose to β -2 of fructose
- C. β -1 of galactose to 4 of glucose
- D. α -1 of glucose to 6 of glucose

Answer: B. α -1 of glucose to β -2 of fructose

Explanation: Sucrose has an $\alpha(1\rightarrow2)\beta$ linkage between glucose C1 and fructose C2 (both anomeric carbons involved).

38. Which of the following is a structural component of plant cell walls?

- A. Glycogen
- B. Starch
- C. Cellulose
- D. Chitin

Answer: C. Cellulose

Explanation: Cellulose is a major structural polysaccharide in plant cell walls, providing rigidity and strength.

39. Glycosaminoglycans (GAGs) were formerly known as:

- A. Polysaccharides
- B. Mucopolysaccharides
- C. Glycoproteins
- D. Proteoglycans

Answer: B. Mucopolysaccharides

Explanation: GAGs were historically called mucopolysaccharides due to their viscous, mucus-like properties.

40. N-linked glycoproteins attach sugars to which amino acid residue?

- A. Serine

- B. Threonine
- C. Asparagine
- D. Lysine

Answer: C. Asparagine

Explanation: N-linked glycosylation attaches oligosaccharides to the amide nitrogen of asparagine residues in proteins.

41. The most common disaccharidase deficiency is:

- A. Sucrase deficiency
- B. Isomaltase deficiency
- C. Lactase deficiency
- D. Maltase deficiency

Answer: C. Lactase deficiency

Explanation: Lactase deficiency (lactose intolerance) is the most common disaccharidase deficiency worldwide.

42. Undigested carbohydrates in the large intestine cause:

- A. Constipation
- B. Osmotic diarrhea
- C. Weight gain
- D. Increased protein synthesis

Answer: B. Osmotic diarrhea

Explanation: Undigested, osmotically active carbohydrates draw water into the colon, causing osmotic diarrhea.

43. Bacterial fermentation of undigested sugars produces which gases?

- A. Oxygen and Nitrogen
- B. CO₂ and H₂
- C. Helium and Neon
- D. Argon

Answer: B. CO₂ and H₂

Explanation: Colonic bacteria ferment undigested sugars, producing gases like carbon dioxide (CO₂) and hydrogen (H₂).

44. A "hydrogen breath test" is used to diagnose:

- A. Diabetes
- B. Protein malnutrition
- C. Carbohydrate malabsorption
- D. Vitamin deficiency

Answer: C. Carbohydrate malabsorption

Explanation: Elevated breath hydrogen indicates carbohydrate malabsorption, as bacteria ferment unabsorbed sugars.

45. Sucrase-isomaltase deficiency is particularly common (10%) in which population?

- A. North Americans

- B. Inuit people
- C. Europeans
- D. Africans

Answer: C. Europeans

Explanation: Sucrase-isomaltase deficiency has a prevalence of about 10% in some European populations.

46. Acarbose is a drug that treats diabetes by inhibiting:

- A. Insulin secretion
- B. α -glucosidase
- C. Glucose transporters
- D. Glucagon

Answer: B. α -glucosidase

Explanation: Acarbose inhibits intestinal α -glucosidase, delaying carbohydrate digestion and lowering postprandial glucose.

47. α -Glucosidase inhibitors have no effect on the digestion of which sugar?

- A. Starch
- B. Sucrose
- C. Lactose
- D. Maltose

Answer: C. Lactose

Explanation: α -Glucosidase inhibitors target α -glycosidic bonds; lactose has a $\beta(1\rightarrow4)$ bond, so it is unaffected.

48. Which condition results in glucose in the urine (glucosuria)?

- A. Galactosemia
- B. Diabetes mellitus
- C. Lactose intolerance
- D. Celiac disease

Answer: B. Diabetes mellitus

Explanation: In diabetes mellitus, hyperglycemia exceeds renal threshold, causing glucose to appear in the urine (glucosuria).

49. Classic galactosemia is caused by a deficiency in:

- A. Galactokinase
- B. Galactose 1-phosphate uridylyltransferase (GALT)
- C. Lactase
- D. Hexokinase

Answer: B. Galactose 1-phosphate uridylyltransferase (GALT)

Explanation: Classic galactosemia is due to GALT deficiency, leading to accumulation of galactose 1-phosphate.

50. The accumulation of which metabolite in the lens causes cataracts in galactosemia?

- A. Sorbitol

- B. Galactitol
- C. Glucose 6-phosphate
- D. UDP-galactose

Answer: B. Galactitol

Explanation: Galactitol accumulates in the lens (via aldose reductase), causing osmotic damage and cataracts.

51. Hereditary Fructose Intolerance (HFI) is due to a deficiency in:

- A. Fructokinase
- B. Aldolase B
- C. Sucrase
- D. GLUT-5

Answer: B. Aldolase B

Explanation: HFI results from a deficiency of aldolase B in the liver, impairing fructose metabolism.

52. Severe hypoglycemia and liver failure in HFI are caused by the accumulation of:

- A. Fructose
- B. Fructose 1-phosphate
- C. Sorbitol
- D. Glucose

Answer: B. Fructose 1-phosphate

Explanation: Accumulation of fructose 1-phosphate traps phosphate, depleting ATP and causing severe hypoglycemia and liver damage.

53. A deficiency of fructokinase causes which benign condition?

- A. Essential fructosuria
- B. Diabetes
- C. Galactosemia
- D. Lactose intolerance

Answer: A. Essential fructosuria

Explanation: Fructokinase deficiency is benign, causing fructose excretion in urine (essential fructosuria) without symptoms.

54. Cataract formation in diabetics is linked to the accumulation of sorbitol in the lens via which enzyme?

- A. Sorbitol dehydrogenase
- B. Aldose reductase
- C. Hexokinase
- D. Glucokinase

Answer: B. Aldose reductase

Explanation: Aldose reductase converts glucose to sorbitol; sorbitol accumulation in the lens leads to osmotic damage and cataracts.

55. Which hormone is released from α -cells when blood glucose drops?

- A. Insulin
- B. Glucagon
- C. Epinephrine
- D. Cortisol

Answer: B. Glucagon

Explanation: Glucagon is released from pancreatic α -cells in response to low blood glucose to stimulate glycogenolysis.

56. Dietary fiber consists of nondigestible carbohydrates and:

- A. Starch
- B. Lignin
- C. Glycogen
- D. Proteins

Answer: B. Lignin

Explanation: Dietary fiber includes nondigestible carbohydrates plus lignin, a non-carbohydrate plant polymer.

57. Soluble fiber has been shown to lower which plasma level?

- A. Blood glucose
- B. LDL cholesterol
- C. Total protein
- D. Vitamin K

Answer: B. LDL cholesterol

Explanation: Soluble fiber (e.g., pectin, beta-glucan) lowers LDL cholesterol by binding bile acids and increasing excretion.

58. Insoluble fiber consists of hemicellulose, lignins, and:

- A. Pectins
- B. Gums
- C. Cellulose
- D. Mucilages

Answer: C. Cellulose

Explanation: Insoluble fiber includes cellulose, hemicellulose, and lignins.

59. Colonic bacteria can metabolize fiber into:

- A. Glucose
- B. Short-chain fatty acids
- C. Amino acids
- D. Vitamins

Answer: B. Short-chain fatty acids

Explanation: Colonic bacteria ferment fiber into short-chain fatty acids (acetate, propionate, butyrate), which are absorbed.

60. Which of the following describes "functional fiber"?

- A. Fiber found only in fruits

- B. Synthetic or extracted fiber with proven health benefits
- C. Fiber that cannot be fermented
- D. Fiber used for energy

Answer: B. Synthetic or extracted fiber with proven health benefits

Explanation: Functional fiber is isolated, extracted, or synthetic fiber that has beneficial physiological effects in humans.

61. What is the major natural sweetener found in honey?

- A. Lactose
- B. Sucrose and free fructose/glucose
- C. Maltose
- D. Galactose

Answer: B. Sucrose and free fructose/glucose

Explanation: Honey is mainly a mixture of sucrose, free fructose, and free glucose.

62. Which enzyme is the first to act on dietary starch in the small intestine?

- A. Salivary amylase
- B. Pancreatic α -amylase
- C. Sucrase
- D. Maltase

Answer: B. Pancreatic α -amylase

Explanation: In the small intestine, pancreatic α -amylase is the first enzyme to act on dietary starch (after salivary amylase is inactivated in the stomach).

63. What happens to salivary α -amylase in the stomach?

- A. It is activated
- B. It is inactivated by high acidity
- C. It digests protein
- D. It turns into pepsin

Answer: B. It is inactivated by high acidity

Explanation: The gastric acid (low pH) denatures and inactivates salivary α -amylase.

64. Lactose synthase is stimulated by which hormone in mammary glands?

- A. Insulin
- B. Glucagon
- C. Prolactin
- D. Estrogen

Answer: C. Prolactin

Explanation: Prolactin stimulates lactose synthase activity in mammary glands during lactation.

65. The B-protein of lactose synthase is also called:

- A. β -galactosyltransferase
- B. α -lactalbumin
- C. Casein
- D. Whey

Answer: B. α -lactalbumin

Explanation: The B-protein component of lactose synthase is α -lactalbumin, which modifies the enzyme's specificity to produce lactose.

66. Which of the following is an intermediate Trisaccharide in starch digestion?

- A. Maltose
- B. Isomaltose
- C. Maltotriose
- D. Lactose

Answer: C. Maltotriose

Explanation: Maltotriose (three glucose units linked by $\alpha(1\rightarrow4)$ bonds) is an intermediate product of starch hydrolysis.

